FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2019

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A boat which has a speed of 6 km/h in still water crosses a river of width 1 km along the shortest possible path in 20 min. The velocity of the river water in km/h is
   (A) 1 (B) 3
   (C) 4 (D) $3\sqrt{3}$

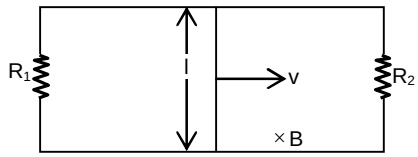
2. A mirror (μ = 3/2) is 10 cm thick. An object is placed 15 cm in front of it. The position of image from the front surface is
   (A) 15 cm (B) 21.67 cm
   (C) 28.34 cm (D) 35 cm

3. A thin prism P_1 with angle 4° and made from glass (μ = 1.54) is combined with another prism P_2 made of another glass of μ = 1.72 to produce dispersion without deviation. The angle of prism P_2 is
   (A) 53.3^o (B) 4°
   (C) 3° (D) 2.6^o

4. A particle starts its motion from rest and moves with constant acceleration for time t_1 and then it retards with constant rate for time t_2 until it comes to rest. Then the ratio of maximum speed and average speed during the complete motion will be
   (A) 2 : 1 (B) 1 : 2
   (C) $t_1 : t_2$ (D) $t_2 : t_1$

Space for Rough work

5. A rectangular loop with a slide wire of length l is kept in a uniform magnetic field as shown in figure. The resistance of slider is R. neglecting self inductance of the loop find the current in the connector during its motion with a velocity v.



(A) $\frac{Blv}{R_1+R_2+R}$ (B) $\frac{Blv(R_1+R_2)}{R+(R_1+R_2)}$

(C) $\frac{Blv(R_1+R_2)}{RR_1+RR_2+R_1R_2}$ (D) $Blv\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)$

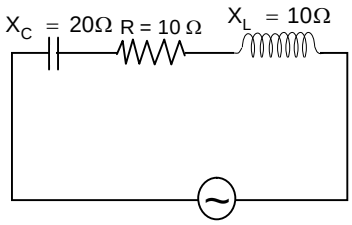
6. A uniform solid brass sphere is rotating with angular speed ω_0 about a diameter. If its temperature is now increased by 100^0C. What will be its new angular speed (Given $\alpha_{brass}=2.0\times10^{-5}\text{per}^0\text{C}$)

(A) $1.1\,\omega_0$ (B) $1.01\,\omega_0$

(C) $0.996\,\omega_0$ (D) $0.824\,\omega_0$

7. When a positively charged conductor is connected to earth,
(A) electrons flow from the earth to the conductor
(B) protons flow from the conductor to the earth
(C) electrons flow from the conductor to the earth
(D) any of A or C may happen

8. In LCR circuit shown in figure, then choose incorrect option
(A) Current will lead the voltage
(B) rms value of current is 20 A
(C) Power factor of the circuit is $\frac{1}{\sqrt{2}}$
(D) Voltage drop across resistance = voltage drop across inductor



Space for Rough work

9. A charged particle of unit mass and unit charge moves with velocity of $\vec{v} = \left(8\hat{i} + 6\hat{j}\right)$ m/ s in a magnetic field of $\vec{B} = 2\hat{k}T$. Choose the correct alternative

(A) The path of the particle may be $x^2 + y^2 - 4x - 21 = 0$

(B) The path of the particle may be $x^2 + y^2 - 4x - 5 = 0$

(C) The path of the particle may be $x^2 + y^2 + 2x - 8 = 0$

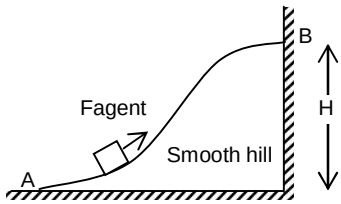
(D) The path of the particle may be $x^2 + y^2 - 2x - 15 = 0$

10. Three charge +q, +q and –q are situated in x-y plane at the point (0 , a) (0 , 0) (0 , –a). Then, the electric potential at a point in first quadrant whose position vector makes an angle θ with y axis and at a distance r(r>>a) from the origin is

(A) $\frac{qa\cos\theta}{2\pi \in_0 r^2}$ (B) $\frac{q}{4\pi \in_0 r}\left(1+\frac{2a\cos\theta}{r}\right)$

(C) $\frac{q}{4\pi \in_0 r}$ (D) $\frac{q}{4\pi \in_0 r}\left(1-\frac{2a\cos\theta}{r}\right)$

11. A external agent moves the block m slowly from A to B, along a smooth hill such that every time he applies the force tangentially. Find the work done by agent in this interval.



(A) $\frac{m^2g^2H^2}{L}$ (B) $\frac{mgH^2}{L}$

(C) mg(H+L) (D) mgH

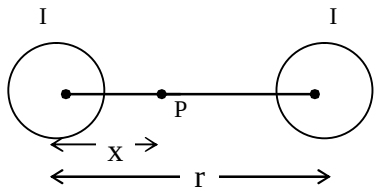
12. A tank has a hole made at its bottom. The time needed to empty the tank from level h_1 to h_2 will be proportional to

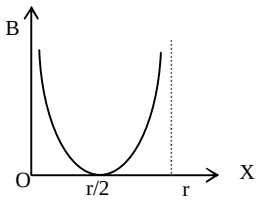
(A) $h_1 + h_2$ (B) $\sqrt{h_1} + \sqrt{h_2}$

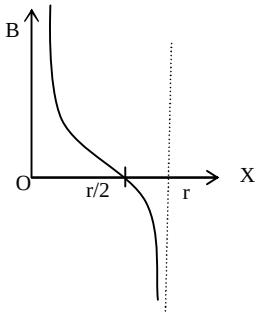
(C) $h_1 - h_2$ (D) $\sqrt{h_1} - \sqrt{h_2}$

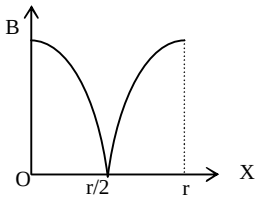
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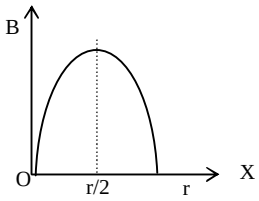
13. Two thin long straight wires are parallel to each other at a separation r apart and they carry current I each along the same directions as shown then induction of magnetic field B, between the wires, varies with x according to



(A)


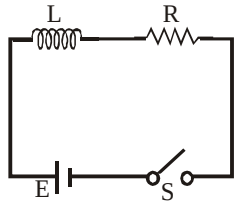
(B)


(C)


(D)


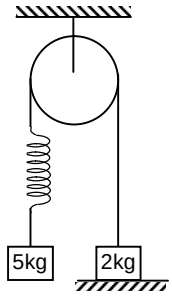
14. One half mole of a monoatomic ideal gas absorbs 1200 J of heat energy while performing 2196 J of work. The temperature of the gas changes by
(A) 160^0C (B) 80^0C
(C) 40^0C (D) -160^0C

15. In the LR circuit as shown in figure, the switch is closed at t = 0. Mark the incorrect statement of the following:
(A)The current increases with time
(B)The current increases with a decreasing rate
(C)The voltage drop across the inductor increases
(D)The voltage drop across the resistor increases



Space for Rough work

16. System shown in figure is released from rest pulley and spring is massless and friction is absent everywhere. The speed of 5 kg block when 2 kg block leaves the contact with ground is (K = 40 N/m & g = 10 m/s^2)



(A) $\sqrt{2}$ m / s (B) $2\sqrt{2}$ m / s

(C) 2 m/s (D) $4\sqrt{2}$ m / s

17. There is a 90^0 bend in a pipe line in which water is flowing at speed v. The cross sectional area of pipe is A. The force exerted by pipe walls on water at bend is

(A) $\sqrt{3}\,A\,v^2\rho$ (B) $\sqrt{2}\,Av^2\rho$

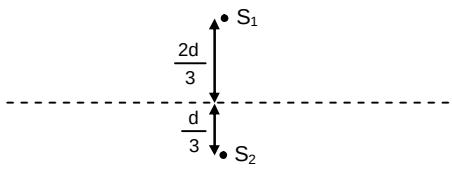
(C) $Av^2\rho$ (D) $\sqrt{5}\,Av\rho$

18. A uniform solid hemisphere of radius r is joined to uniform solid right circular cone of base of radius r. Both have same density. The centre of mass of the composite solid lies on the common face. The height (h) of the cone is

(A) 2r (B) $\sqrt{3}\,r$

(C) 3r (D) $r\sqrt{6}$

19. Two coherent point sources of frequency ($f = \frac{10v}{d}$ where v is speed of light) are placed at a distance d apart as shown in figure. The receiver is free to move along the dotted line shown in the figure. Find total number of maxima observed by receiver.



(A) 6 (B) 7

(C) 5 (D) 8

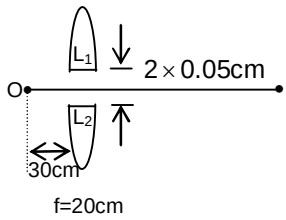
Space for Rough work

20. A hydrogen atom and a Li^{++} ions are both in the second excited state. If ℓ_H and ℓ_{Li} are their respective electronic angular momenta, and E_H and E_{Li} their respective energies, then

(A) $\ell_H > \ell_{Li}$ and $|E_H| > |E_{Li}|$ (B) $\ell_H = \ell_{Li}$ and $|E_H| < |E_{Li}|$

(C) $\ell_H = \ell_{Li}$ and $|E_H| > |E_{Li}|$ (D) $\ell_H < \ell_{Li}$ and $|E_H| < |E_{Li}|$

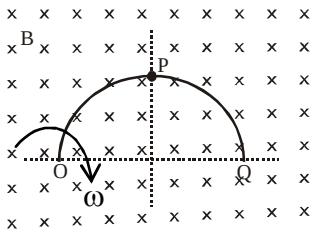
21. A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each of which is displaced by 0.0005 m as shown in figure. Find the distance between the images (in mm)



(A) 1 (B) 2

(C) 3 (D) 4

22. The figure shows a semi-circular conducting ring of radius a, which is rotated with constant angular velocity ω about an axis passing through O and parallel to field B as shown. Identify the correct value of potential difference



(A) $V_P - V_Q = \frac{1}{2}B\omega a^2$

(B) $V_Q - V_O = 2B\omega a^2$

(C) $V_P - V_O = \frac{1}{4}B\omega a^2$

(D) The maximum value of $|V_P - V_O| = \frac{\pi a^2 \omega B}{2}$

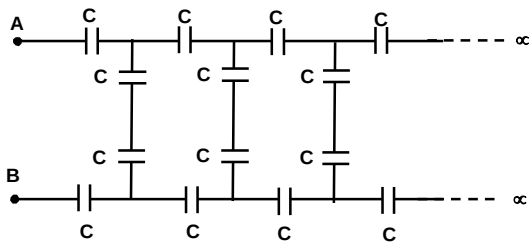
23. A train of mass M is moving on a circular track of radius R with constant speed v. The length of train is half the perimeter of track. The linear momentum of the train will be

(A) 0 (B) $2M\nu/\pi$

(C) MvR (D) Mv

Space for Rough work

24. The equivalent capacitance of the shown infinite network across A and B, if each capacitor has a capacitance of C, is



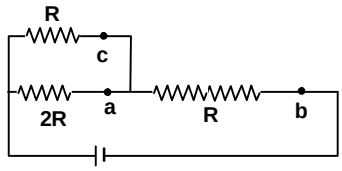
(A) $\left(\frac{\sqrt{3}-1}{4}\right)C$

(B) $\left(\frac{\sqrt{3}+1}{4}\right)C$

(C) $\left(\frac{\sqrt{5}-1}{4}\right)C$

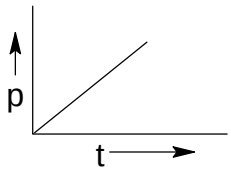
(D) None of these

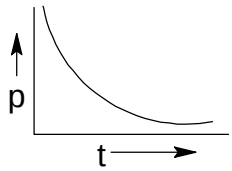
25. Referring to the shown circuit, the current will be minimum in

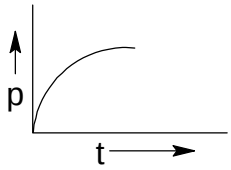


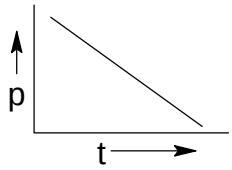
(A) a
(B) b
(C) c
(D) same in all the branches

26. A soap bubble is blown slowly at the end of a tube by a pump supplying air at a constant rate. Which one of the following graphs represents the correct variation of the excess of pressure inside the bubble with time?

(A) 

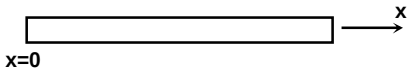
(B) 

(C) 

(D) 

Space for Rough work

27. If along a uniform rod of length ℓ carrying current I, the voltage V changes with position x along the length of the rod such that dV/dx = – k, where k is a positive number, then the resistance of the rod is



(A) $k\ell/2I$ (B) $k\ell/I$

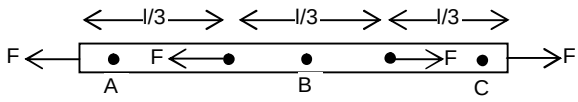
(C) $I/k\ell$ (D) $k\ell I$

28. A cylindrical hall has a horizontal smooth floor. A ball is projected along the floor from A point on the wall in a direction making an angle θ with the radius through the point the ball returns back to the initial point after two impacts with the wall. If the coefficient of restitution is e then $\tan^2\theta$ will be

(A) $\frac{1+e+e^2}{e^3}$ (B) $\frac{1+e}{e^2}$

(C) $\frac{e^2}{1+e}$ (D) $\frac{e^3}{1+e+e^2}$

29. Four persons are holding a rope of length ℓ at position as shown in figure. All of them are applying equal force F in the direction as shown in figure. Then the tension



Statement 1 – at point A will be F
Statement 2 – at point B will be 2F
Statement 3 – at point C will be F

(A) only statement 1 & 2 is true (B) only statement 2 is true

(C) all the three statements are true (D) none of the statements are true

30. The mass of a body is measured as 10.1 kg. The possible percentage error in the measurement is

(A) $\pm$ 1% (B) $\pm$ 0.1%

(C) $\pm$ 10% (D) $\pm$ 0.01%

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Among the following oxoacids, the correct decreasing order of acid strength is:
(A) $HClO_4 > HClO_3 > HClO_2 > HOCl$ (B) $HClO_2 > HClO_4 > HClO_3 > HOCl$
(C) $HOCl > HClO_2 > HClO_3 > HClO_4$ (D) $HClO_4 > HOCl > HClO_2 > HClO_3$

32. In which of the following species $p\pi$-$d\pi$ bond is present but $p\pi$-$p\pi$ bond is absent?
(A) SiH_4 (B) CS_2
(C) SO_2 (D) SO_2Cl_2

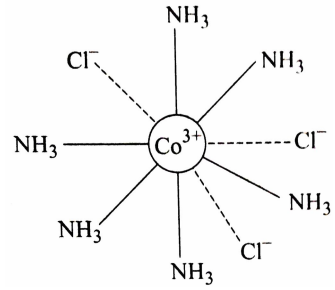
33. Nodal plane of all the π-bonds is in same plane:
(A) C_6H_6 (B) CO_2
(C) $H_2C = C = CH_2$ (D) N_2

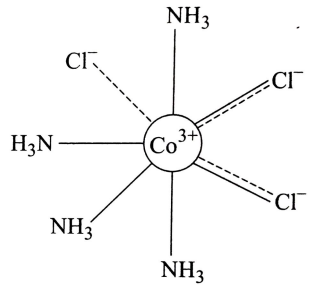
34. In which of the following pairs both are monobasic, proton donor acid when dissolve in water?
(A) H_3PO_2, H_3PO_3 (B) H_3BO_3, H_3PO_2
(C) $HClO_3$, H_3BO_3 (D) H_3PO_2, $HClO_4$

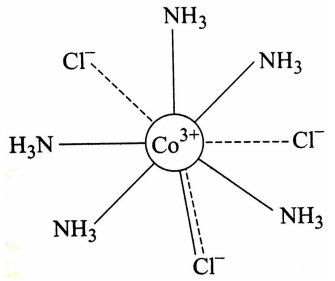
35. Which of the following paramagnetic complex of metal in its +2 oxidation state shows geometrical isomerism?
(A) $\left[Ni(en)_3\right]Cl_2$ (B) $\left[Co(H_2O)_5NO_2\right]Cl$
(C) $\left[Pt(NH_3)_2Cl_2\right]$ (D) $\left[Co(NH_3)_4(H_2O)_2\right]Cl_2$

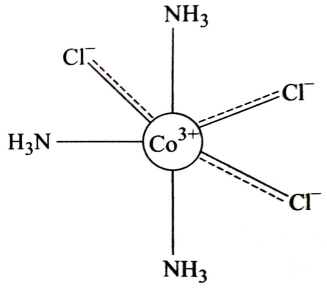
Space for Rough work

36. Which of the following complexes (Werner presentation) have minimum electrical conductance in aqueous solution?

(A) 

(B) 

(C) 

(D) 

37. Sulphide ores are common for the metals:
(A) Ag, Cu and Pb (B) Ag, Cu and Sn
(C) Ag, Mg and Pb (D) Al, Cu and Pb

38. Which of the following bicarbonates does not exist in solid state?
(A) $LiHCO_3$ (B) $NaHCO_3$
(C) $KHCO_3$ (D) $RbHCO_3$

39. Which of the following does not give gaseous product on treatment with concentrated HNO_3?
(A) $BaCO_3$ (B) $NaNO_2$
(C) $Ca_3(PO_4)_2$ (D) $CaBr_2$

40. Hard water does not contain:
(A) Ca^{2+} with HCO_3^- (B) Mg^{2+} with SO_4^{2-}
(C) Mg^{2+} with Cl^- (D) Ca^{2+} with CO_3^{2-}

Space for Rough work

41. 100 mL of H_2SO_4 solution having molarity 1 M and density 1.5 g/mL is mixed with 400 mL of water. Calculate final molarity of H_2SO_4 solution, if final density is 1.25 g/mL:
(A) 4.4 M (B) 0.145 M
(C) 0.52 M (D) 0.227 M

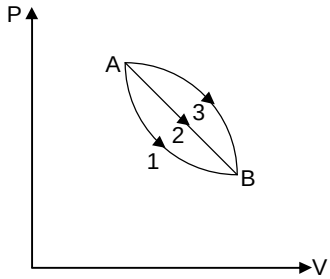
42. When $K_2Cr_2O_7$ is converted to K_2CrO_4, the change in the oxidation state of chromium is:
(A) 0 (B) 6
(C) 4 (D) 3

43. The mass of an electron is m, charge is e and it is accelerated from rest through a potential difference of V volts. The velocity acquired by electron will be:
(A) $\sqrt{\frac{V}{m}}$ (B) $\sqrt{\frac{eV}{m}}$
(C) $\sqrt{\frac{2eV}{m}}$ (D) zero

44. Wavelength for high energy EMR transition in H-atom is 91 nm. What energy is needed for this transition?
(A) 1.36 eV (B) 1240 eV
(C) 13 eV (D) 13.6 eV

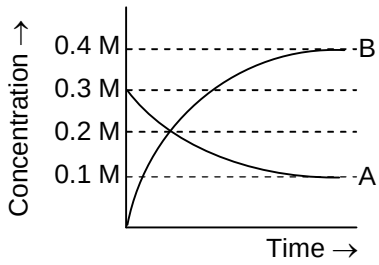
45. An ideal gaseous mixture of ethane (C_2H_6) and ethene (C_2H_4) occupies 28 litre at 1 atm and 273 K. the mixture reacts completely with 128 gm O_2 to produce CO_2 and H_2O. Mole fraction at C_2H_4 in the mixture is:
(A) 0.6 (B) 0.4
(C) 0.5 (D) 0.8

46. A given mass of gas expands reversibly from the state A to the state B by three paths 1, 2 and 3 as shown in the figure. If w_1, w_2 and w_3 respectively be the work done by the gas along three paths then:
(A) $w_1 > w_2 > w_3$
(B) $w_1 < w_2 < w_3$
(C) $w_1 = w_2 = w_3$
(D) $w_2 > w_3 > w_1$



Space for Rough work

47. The figure shows the change in concentration of species A and B as a function of time.The equilibrium constant K_c for the reaction $A(g) \rightleftharpoons 2B(g)$ is:



(A) $K_c > 1$
(B) $K < 1$
(C) $K = 1$
(D) data insufficient

48. What concentration of $HCOO^-$ is present in a solution of 0.01 M HCOOH ($K_a = 1.8 \times 10^{-4}$) and 0.01 M HCl?
(A) 1.8×10^{-3} (B) 10^{-2}
(C) 1.8×10^{-4} (D) 10^{-4}

49. In the following reaction, how is the rate of appearance of the underlined product related to the rate of disappearance of the underlined reactant

$$\underline{BrO_3^-(aq)} + 5Br^-(aq) + 6H^+(aq) \longrightarrow \underline{3Br_2(\ell)} + 3H_2O(aq)$$

(A) $-\frac{d[BrO_3^-]}{dt} = \frac{d[Br_2]}{dt}$ (B) $-\frac{1}{3}\frac{d[BrO_3^-]}{dt} = \frac{d[Br_2]}{dt}$
(C) $\frac{-d[BrO_3^-]}{dt} = \frac{1}{3}\frac{d[Br_2]}{dt}$ (D) None of these

50. The cell reaction $Hg_2Cl_2(s) + Cu(s) \longrightarrow Cu^{2+}(aq) + 2Cl^-(aq) + 2Hg(\ell)$, is best represented by:
(A) $Cu(s)\,|\,Cu^{2+}(aq)\,||\,Hg_2Cl_2(s)\,|\,Hg(\ell)$
(B) $Cu(s)\,|\,Cu^{2+}(aq)\,||\,Hg(\ell)\,|\,HgCl_2(s)$
(C) $Cu(s)\,|\,Cu^{2+}(aq)\,||\,Cl^-(aq)\,|\,Hg_2Cl_2(s)\,|\,Hg(\ell)\,|\,Pt(s)$
(D) $Hg_2Cl_2(s)\,|\,Cl^-(aq)\,||\,Cu^{2+}(aq)\,|\,Cu(s)$

Space for Rough work

51. The degree of dissociation of an electrolyte is α and its van't Hoff factor is i. The number of ions obtained by complete dissociation of 1 molecule of the electrolyte is:

(A) $\frac{i+\alpha-1}{\alpha}$ (B) $i-\alpha-1$

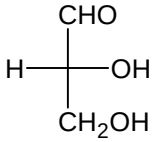
(C) $\frac{i-1}{\alpha}$ (D) $\frac{i+1+\alpha}{1-\alpha}$

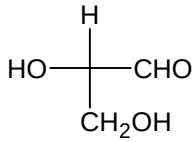
52. The co-ordination number of a metal crystallising in a hexagonal close-packed structure is:

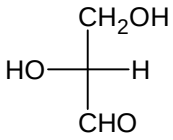
(A) 12 (B) 4

(C) 8 (D) 6

53. Which is the correct structure of D-glyceraldehyde?

(A) 

(B) 

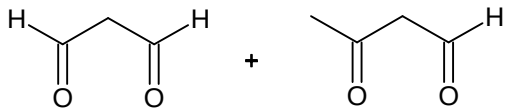
(C) 

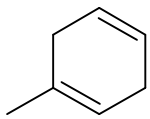
(D) All

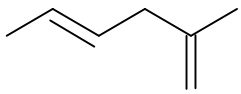
54. Which alkene on oxidation with acidic $KMnO_4$ gives only acetic acid?

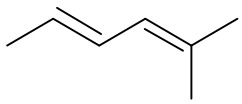
(A) $CH_3 = CH - CH_3$ (B) $CH_3 - CH = CH - CH_3$

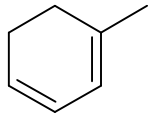
(C) Ethylene (D) Pentene – 2

55. Compound (X) $\xrightarrow{O_3, Zn-H_2O}$ 

(A) 

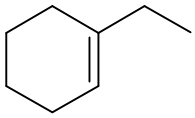
(B) 

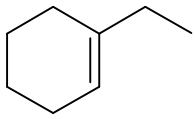
(C) 

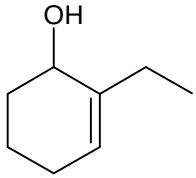
(D) 

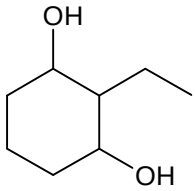
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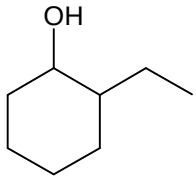
56. Vinylic halides are unreactive towards nucleophilic substitution because of the following except:
(A) C-halogen bond is strong
(B) The halogen is bonded to sp^2 carbon
(C) A double bond character is developed in the carbon-halogen bond by resonance
(D) Halide ions are not good leaving groups

57.  $\xrightarrow{SeO_2}$ A, A is :

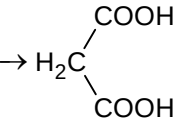
(A) 

(B) 

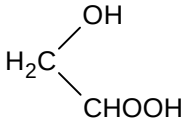
(C) 

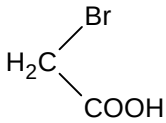
(D) 

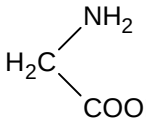
58. Which will give silver mirror test with Tollen's reagent?
(A) C_6H_5CHO (B) $CH_3 - CHO$
(C) HCOOH (D) All of these

59. In the reaction A $\xrightarrow{KCN}$ B $\xrightarrow[\text{reflux}]{H_3O^{\oplus}}$ 

Identify A:

(A) 

(B) 

(C) 

(D) All of these

60. PhCOOH and Ph – CH_3 can be separated by:
(A) $NaHCO_3$ (B) aq. $NaHCO_3$ + n-Hexane
(C) H_2O (D) n-Hexane

Space for Rough work

Mathematics PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $f(x) = [9^x - 3^x + 1] \forall x \in (-\infty, 1)$, then range of f(x) is ([·] denotes the greatest integer function)
(A) {0, 1, 2, 3, 4, 5, 6} (B) {0, 1, 2, 3, 4, 5, 6, 7}
(C) {1, 2, 3, 4, 5, 6} (D) {1, 2, 3, 4, 5, 6, 7}

62. If α and β are the roots of the equation $ax^2 + bx + c = 0$ such that $\beta < \alpha < 0$, then the quadratic equation whose roots are $|\alpha|$, $|\beta|$ is given by :
(A) $|a|x^2 + |b|x + |c| = 0$ (B) $ax^2 - b|x| + c = 0$
(C) $|a|x^2 - |b|x + |c| = 0$ (D) $a|x|^2 + b|x| + c = 0$

63. if $\alpha_0, \alpha_1, \alpha_2, \ldots\ldots, \alpha_{n-1}$, be the n, n^{th} roots of the unity, then the value of $\sum_{i=0}^{n-1} \frac{\alpha_i}{(3-\alpha_i)}$ is equal to :
(A) $\frac{n}{3^n - 1}$ (B) $\frac{n-1}{3^n - 1}$
(C) $\frac{n+1}{3^n - 1}$ (D) $\frac{n+2}{3^n - 1}$

64. If the ratio of A.M. between two positive real numbers a and b to their H.M. is m : n; then a : b is equal to :
(A) $\frac{\sqrt{(m-n)} + \sqrt{n}}{\sqrt{(m-n)} - \sqrt{n}}$ (B) $\frac{\sqrt{n} + \sqrt{(m-n)}}{\sqrt{n} - \sqrt{(m-n)}}$
(C) $\frac{\sqrt{m} + \sqrt{(m-n)}}{\sqrt{m} - \sqrt{(m-n)}}$ (D) $\frac{\sqrt{(m-n)} + \sqrt{m}}{\sqrt{(m-n)} - \sqrt{m}}$

Space for Rough work

65. Given that $(1+\sqrt{(1+x)}\tan y = 1+(\sqrt{1-x})$, then sin 4y is equal to:
(A) 4x (B) 2x
(C) x (D) none of these

66. Number of solutions of the equations $y = \frac{1}{3}[\sin x + [\sin x + [\sin x]]]$ and $[y + [y]] = 2\cos x$, where [·] denotes the greatest integer function is :
(A) 0 (B) 1
(C) 2 (D) infinite

67. Let f(x + y) = f(x) · f(y) for all x and y and f(1) = 2. If in a triangle ABC, a = f(3), b = f(1) + f(3), c = f(2) + f(3) , then 2A is equal to :
(A) C (B) 2C
(C) 3C (D) 5C

68. If $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$, then the two curves y = cos x and y = sin 3x intersect at
(A) $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ and $\left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right)$ (B) $\left(-\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{\pi}{8}, \cos\frac{\pi}{8}\right)$
(C) $\left(\frac{\pi}{4}, -\frac{1}{\sqrt{2}}\right)$ and $\left(\frac{\pi}{8}, -\cos\frac{\pi}{8}\right)$ (D) $\left(-\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$

69. The number of ways in which 9 identical balls can be placed in three identical boxes is :
(A) 55 (B) $\frac{9!}{(3!)^4}$
(C) $\frac{9!}{(3!)^3}$ (D) 12

Space for Rough work

70. If C_r stands for ${}^nC_r$, then the coefficient of $\lambda^n\mu^n$ in the expansion of $[(1+\lambda)(1+\mu)(\lambda+\mu)]^n$ is :

(A) $\sum_{r=0}^{n} C_r^2$ (B) $\sum_{r=0}^{n} C_{r+2}^2$

(C) $\sum_{r=0}^{n} C_{r+3}^2$ (D) $\sum_{r=0}^{n} C_r^3$

71. A piece of wire of length 4L is bent at random, to form a rectangle. The probability that its area is at most $\frac{L^2}{4}$ is :

(A) 1/4 (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{2-\sqrt{3}}{2}$ (D) $4-\sqrt{3}$

72. The number of distinct values of a 2×2 determinant whose entries are from the set {–1, 0, 1} is:

(A) 3 (B) 4

(C) 5 (D) 6

73. If A is non-singular matrix, then $\det\cdot(A^{-1})$ is equal to :

(A) $\det\cdot\left(\frac{1}{A^2}\right)$ (B) $\frac{1}{\det\cdot(A^2)}$

(C) $\det\cdot\left(\frac{1}{A}\right)$ (D) $\frac{1}{\det\cdot(A)}$

74. The four sides of a quadrilateral are given by the equation $xy(x-2)(y-3)=0$. The equation of the line parallel to $x-4y=0$ that divides the quadrilateral in two equal areas is :

(A) $x-4y-5=0$ (B) $x-4y+5=0$

(C) $x-4y-1=0$ (D) $x-4y+1=0$

Space for Rough work

75. The equation of image of pair of lines $y = |x-1|$ in y-axis is :

(A) $x^2 + y^2 + 2x + 1 = 0$ (B) $x^2 - y^2 + 2x - 1 = 0$

(C) $x^2 - y^2 + 2x + 1 = 0$ (D) none of these

76. The equation of the image of the circle $(x - 3)^2 + (y - 2)^2 = 1$ by the mirror $x + y = 19$ is :

(A) $(x - 14)^2 + (y - 13)^2 = 1$ (B) $(x - 15)^2 + (y - 14)^2 = 1$

(C) $(x - 16)^2 + (y - 15)^2 = 1$ (D) $(x - 17)^2 + (y - 16)^2 = 1$

77. The point ([P + 1], [P]), (where [·] denotes the greatest integer function) lying inside the region bounded by the circle $x^2 + y^2 - 2x - 15 = 0$ and $x^2 + y^2 - 2x - 7 = 0$, then :

(A) $P \in [-1,0) \cup [0, 1) \cup [1, 2)$ (B) $P \in [-1,2) - \{0,1\}$

(C) $P \in (-1,2)$ (D) none of the above

78. Let α be the angle which a tangent to the parabola $y^2 = 4ax$ makes with its axis, the distance between the tangent and a parallel normal will be :

(A) $a\sin^2\alpha\cos^2\alpha$ (B) $a\operatorname{cosec}\alpha\sec^2\alpha$

(C) $a\tan^2\alpha$ (D) $a\cos^2\alpha$

79. The point, at shortest distance from the line $x + y = 7$ and lying on an ellipse $x^2 + 2y^2 = 6$, has co-ordinates :

(A) $(\sqrt{2},\sqrt{2})$ (B) $(0,\sqrt{3})$

(C) (2,1) (D) $\left(\sqrt{5},\frac{1}{\sqrt{2}}\right)$

80. The normal at P to a hyperbola of eccentricity e, intersects its transverse and conjugate axes at L and M respectively. If locus of the mid-point of LM is hyperbola, then eccentricity of the hyperbola is :

(A) $\left(\frac{e+1}{e-1}\right)$ (B) $\frac{e}{\sqrt{(e^2-1)}}$

(C) e (D) none of these

Space for Rough work

81. If x is a real number in [0, 1] then the value of $f(x) = \lim_{m\to\infty}\lim_{n\to\infty}\{1+\cos^{2m}(n!\pi x)\}$ is given by :

(A) 2 or 1 according as x is rational or irrational
(B) 1 or 2 according as x is rational or irrational
(C) 1 for all x
(D) 2 or 1 for all x

82. If $\sqrt{(x+y)}+\sqrt{(y-x)}=a$, then $\frac{d^2y}{dx^2}$ equals :

(A) 2 / a
(B) $-2 / a^2$
(C) $2 / a^2$
(D) none of these

83. The angle of intersection of curves $Y=\left[|\sin x|+|\cos x|\right]$ and $x^2 + y^2 = 5$, where [·] denotes the greatest integer function, is :

(A) $\tan^{-1} 2$
(B) $\tan^{-1}\left(\frac{1}{2}\right)$
(C) $\tan^{-1}(\sqrt{2})$
(D) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

84. $\int x^{-2/3}(1+x^{1/2})^{-5/3}\,dx$ is equal to :

(A) $3(1+x^{-1/2})^{-1/3}+C$
(B) $3(1+x^{-1/2})^{-2/3}+C$
(C) $3(1+x^{1/2})^{-2/3}+C$
(D) none of these

85. If [x] denotes the greatest integer less than or equal to x then $\int_0^{\infty}\left[\frac{2}{e^x}\right]dx$ is equal to :

(A) $\log_e 2$
(B) e^2
(C) 0
(D) $\frac{2}{e}$

Space for Rough work

86. Value of $\int_2^3 \frac{dx}{\sqrt{(1+x^3)}}$ is :

(A) less than 1 (B) greater than 2
(C) lies between 3 and 4 (D) none of these

87. The slope of the tangent to a curve y = f(x) at (x, f(x)) is 2x + 1. If the curve passes through the point (1, 2), then the area of the region bounded by the curve, the x-axis and the line x = 1 is:

(A) $\frac{5}{6}$ sq. units (B) $\frac{6}{5}$ sq. units
(C) $\frac{1}{6}$ sq. units (D) 6 sq. units

88. The curve for which sum of the reciprocals of the radius vector and the polar subtangent is k, is equal to :

(A) $kr = c\, r\, e^{\theta} + 1$ (B) $k\, r^2 = c\, e^{\theta} + 1$
(C) $r = k\, r\, e^{\theta} + c$ (D) $k^2\, r^2 = k\, e^{\theta} + c$

89. The three planes 4y + 6z = 5; 2x + 3y + 5z = 5 6x + 5y + 9z = 10 :

(A) meet in a point (B) have a line in common
(C) form a triangular prism (D) none of these

90. The condition for the lines x = az + b, y = cz + d and $x = a_1z + b_1$, $y = c_1z + d_1$ to be perpendicular is :

(A) $ac_1 + a_1c = 1$ (B) $aa_1 + cc_1 + 1 = 0$
(C) $bc_1 + b_1c + 1 = 0$ (D) none of the above

Space for Rough work